

Triagem Tripsina Anticarsia e
Spodoptera GORE4

Preparar moléculas

- Receptor/ligand
- <https://server.poissonboltzmann.org/pdb2pqr>
- Ajuste de pH para 8,2
- Campo de força amber
- Exportar arquivo .pqr
- Abrir no pymol e exportar .pdb

Haddock

- Carregar no Haddock, escolher resíduos ativo e executar default

Dinâmica Molecular

- 1. PREPARE_PH — roda pdb2pqr no receptor e ligante, determinando o estado de protonação correto a pH 8.2 via PropKa (HIS, ASP, GLU, etc.)
- 2. PREPARE_COMPLEX — junta receptor + ligante num único PDB, renomeia resíduos para o campo de força AMBER
- 3. TOPOLOGY — pdb2gmx gera a topologia AMBER99SB-ILDN + água TIP3P
- 4. BOX_SOLVATE_IONS — caixa dodecaédrica (margem 1.2 nm), solvatação, neutralização + NaCl 0.15 M
- 5. MINIMIZATION — minimização de energia (steepest descent)
- 6. NVT — equilibração de temperatura (300 K, 100 ps)
- 7. NPT — equilibração de pressão (1 bar, 100 ps)
- 8. PRODUCTION — dinâmica molecular de 100 ns na GPU (RTX 5070 Ti)
- 9. POSTPROCESS — correção de periodicidade, fitting da trajetória
- 10. ANALYSES — RMSD backbone, RMSD ligante, RMSF, raio de giro, contatos, pontes de hidrogênio
- 11. PLOT — painel de 6 figuras com todas as métricas (título: ACR157-GORE4 - DM 100 ns @ pH 8.2)
- 12. MMGBSA — cálculo de energia de ligação ΔG (MM-GBSA) + painel final de 8 figuras incluindo componentes VdW/Elet/GB/SA

Resultados Spodoptera

- XP273-GORE4-NEW

Cluster 1

HADDOCK score	-43.1 +/- 2.1
Cluster size	39
RMSD from the overall lowest-energy structure	0.5 +/- 0.3
Van der Waals energy	-17.5 +/- 1.0
Electrostatic energy	-57.2 +/- 3.5
Desolvation energy	-14.2 +/- 1.5
Restraints violation energy	0.2 +/- 0.2
Buried Surface Area	725.9 +/- 18.9
Z-Score	-1.3

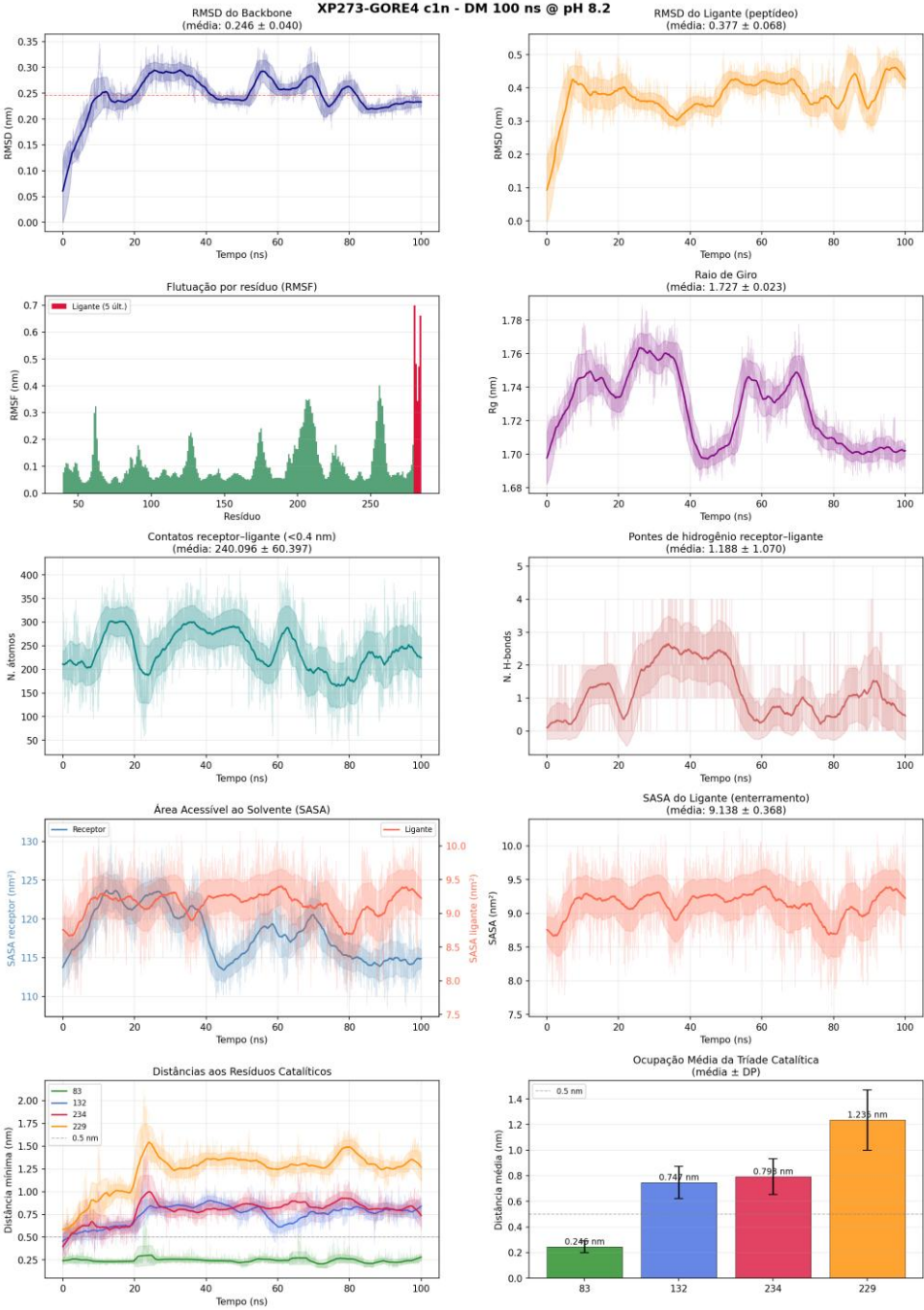
Cluster 6

HADDOCK score	-42.5 +/- 0.8
Cluster size	12
RMSD from the overall lowest-energy structure	0.7 +/- 0.3
Van der Waals energy	-24.5 +/- 2.0
Electrostatic energy	-18.6 +/- 13.5
Desolvation energy	-15.7 +/- 0.8
Restraints violation energy	14.7 +/- 12.5
Buried Surface Area	781.4 +/- 3.4
Z-Score	-1.2

Cluster 2

HADDOCK score	-40.7 +/- 1.8
Cluster size	27
RMSD from the overall lowest-energy structure	2.3 +/- 0.1
Van der Waals energy	-21.2 +/- 3.8
Electrostatic energy	-68.9 +/- 30.7
Desolvation energy	-8.4 +/- 1.4
Restraints violation energy	27.1 +/- 25.8
Buried Surface Area	653.4 +/- 9.2
Z-Score	-0.9

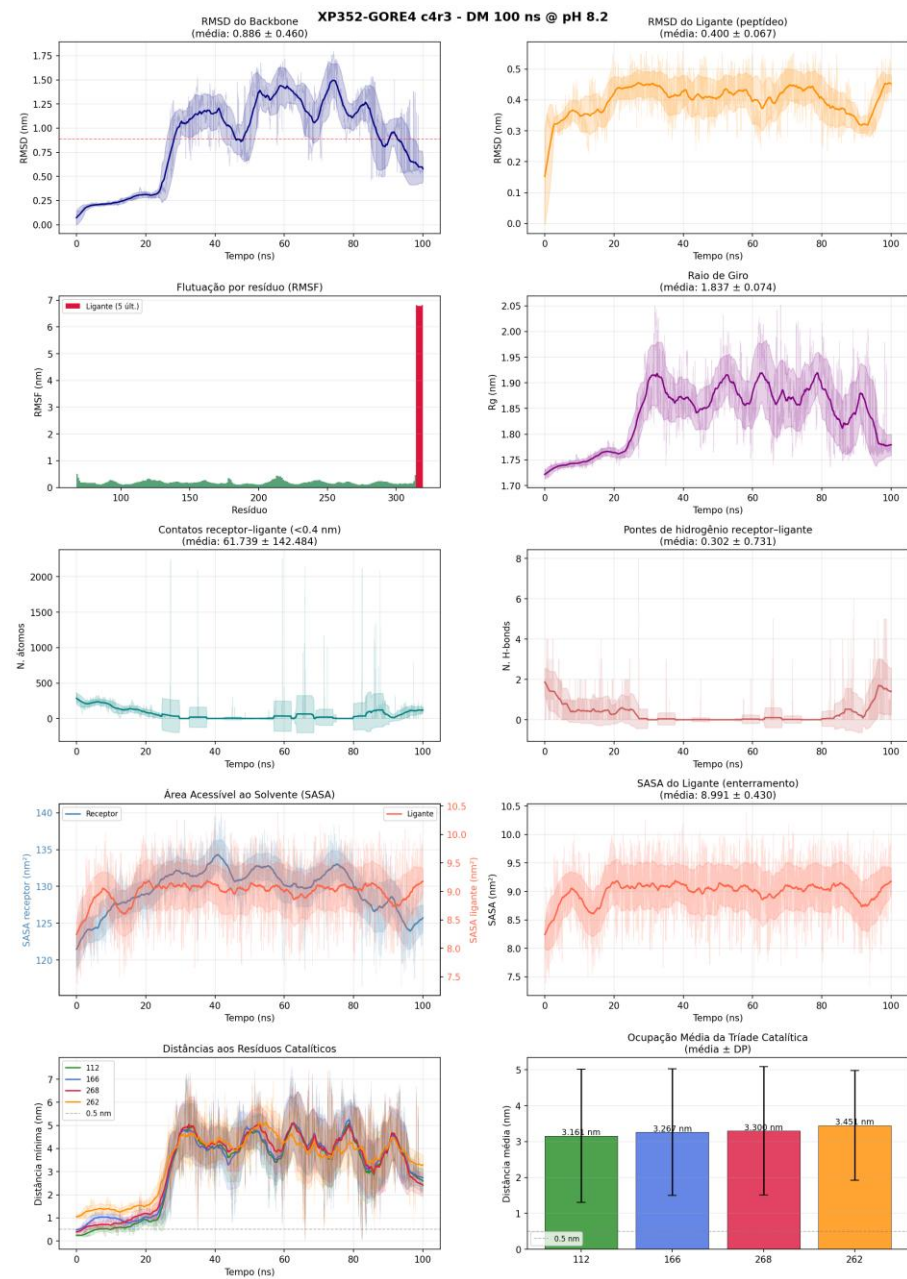
MD



- XP352-GORE4-NEW

Cluster 4		Cluster 10		Cluster 2	
HADDOCK score	-68.2 +/- 3.9	HADDOCK score	-53.9 +/- 4.7	HADDOCK score	-53.1 +/- 1.0
Cluster size	12	Cluster size	6	Cluster size	17
RMSD from the overall lowest-energy structure	0.1 +/- 0.1	RMSD from the overall lowest-energy structure	0.5 +/- 0.2	RMSD from the overall lowest-energy structure	0.7 +/- 0.0
Van der Waals energy	-26.5 +/- 3.4	Van der Waals energy	-20.9 +/- 3.8	Van der Waals energy	-27.7 +/- 1.3
Electrostatic energy	-167.3 +/- 7.6	Electrostatic energy	-131.1 +/- 13.9	Electrostatic energy	-89.0 +/- 4.8
Desolvation energy	-10.8 +/- 0.7	Desolvation energy	-9.4 +/- 0.8	Desolvation energy	-9.9 +/- 0.3
Restraints violation energy	25.2 +/- 2.7	Restraints violation energy	25.3 +/- 18.6	Restraints violation energy	23.1 +/- 13.2
Buried Surface Area	895.9 +/- 15.2	Buried Surface Area	834.0 +/- 29.3	Buried Surface Area	859.7 +/- 9.8
Z-Score	-2.5	Z-Score	-0.5	Z-Score	-0.4

MD



- QCL36-GORE4-NEW

Cluster 1

HADDOCK score	-45.7 +/- 0.7
Cluster size	89
RMSD from the overall lowest-energy structure	0.8 +/- 0.0
Van der Waals energy	-28.1 +/- 2.0
Electrostatic energy	-70.3 +/- 5.2
Desolvation energy	-3.5 +/- 0.3
Restraints violation energy	0.1 +/- 0.2
Buried Surface Area	805.6 +/- 33.0
Z-Score	-1.2

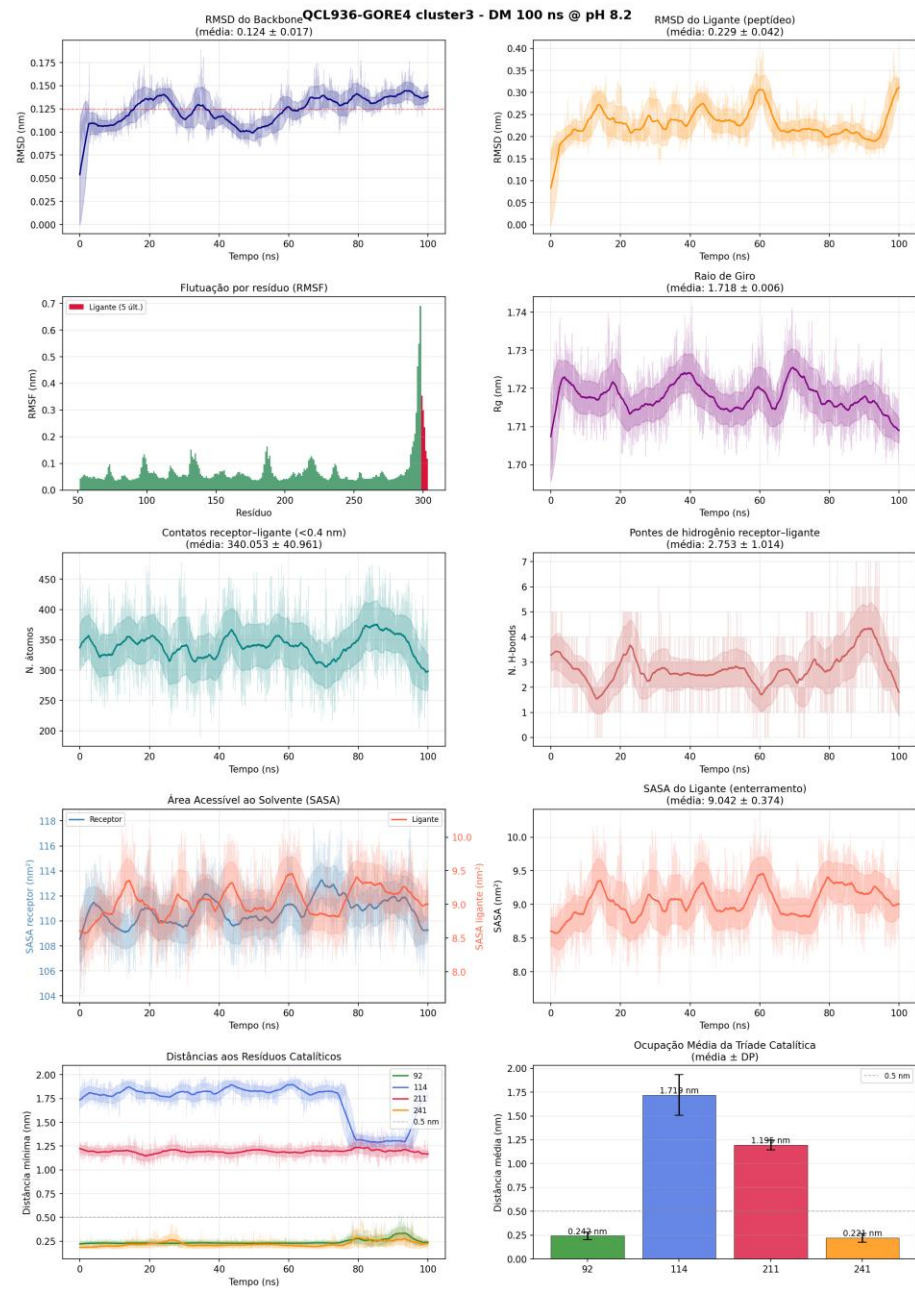
Cluster 3

HADDOCK score	-44.7 +/- 4.5
Cluster size	31
RMSD from the overall lowest-energy structure	0.3 +/- 0.2
Van der Waals energy	-26.9 +/- 2.5
Electrostatic energy	-92.3 +/- 19.7
Desolvation energy	0.5 +/- 0.6
Restraints violation energy	2.1 +/- 1.7
Buried Surface Area	767.9 +/- 49.8
Z-Score	-1.0

Cluster 2

HADDOCK score	-41.5 +/- 1.5
Cluster size	43
RMSD from the overall lowest-energy structure	1.0 +/- 0.1
Van der Waals energy	-26.4 +/- 2.2
Electrostatic energy	-54.1 +/- 2.9
Desolvation energy	-4.4 +/- 0.9
Restraints violation energy	0.2 +/- 0.2
Buried Surface Area	769.6 +/- 44.0
Z-Score	-0.4

MD



- ACR157-GORE4

Cluster 1

HADDOCK score	-46.1 +/- 0.5
Cluster size	90
RMSD from the overall lowest-energy structure	0.8 +/- 0.0
Van der Waals energy	-24.6 +/- 1.3
Electrostatic energy	-94.7 +/- 1.8
Desolvation energy	-3.2 +/- 0.4
Restraints violation energy	6.3 +/- 6.8
Buried Surface Area	758.8 +/- 13.3
Z-Score	-1.5

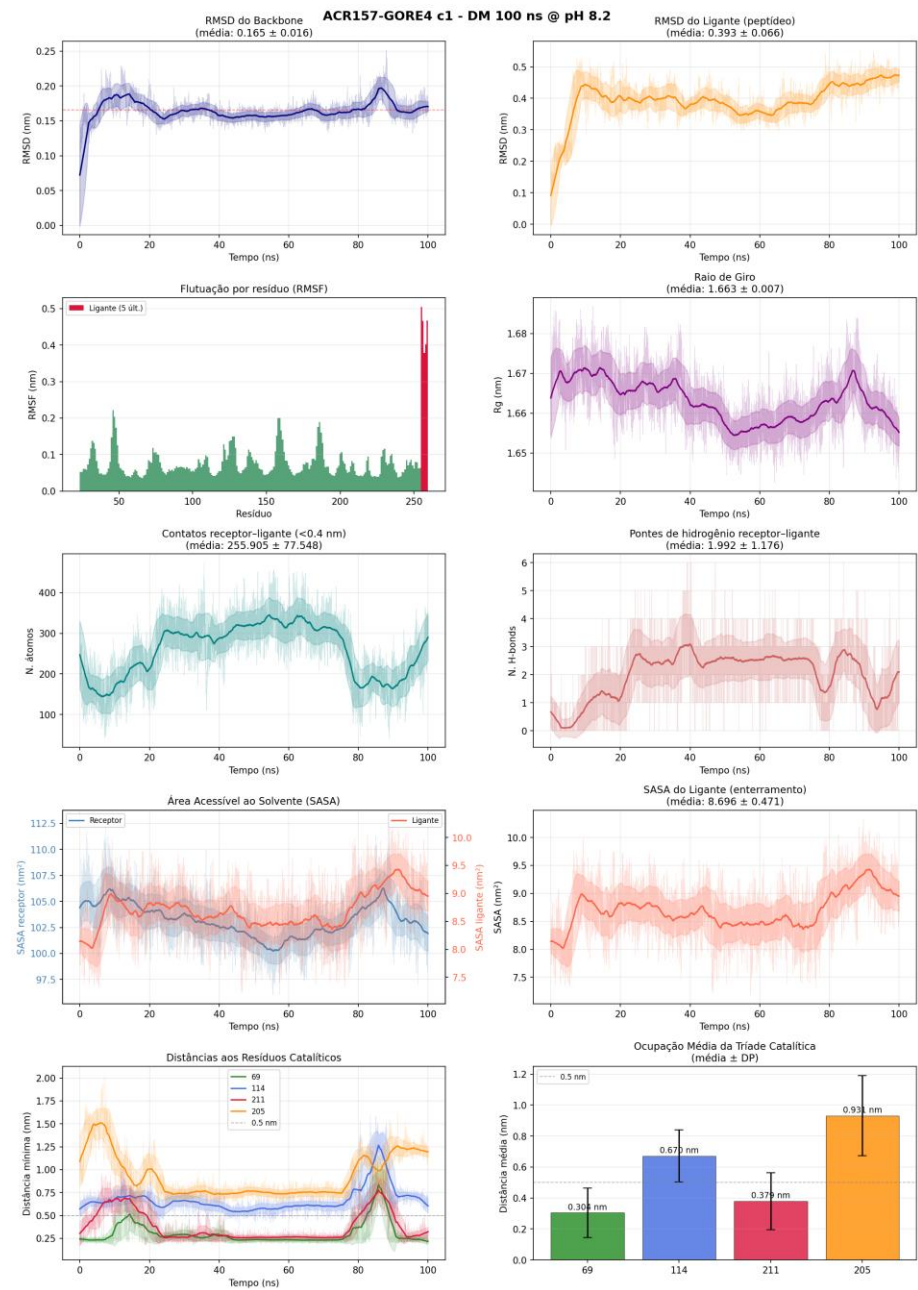
Cluster 2

HADDOCK score	-42.5 +/- 1.9
Cluster size	50
RMSD from the overall lowest-energy structure	0.9 +/- 0.1
Van der Waals energy	-16.5 +/- 2.9
Electrostatic energy	-110.8 +/- 24.1
Desolvation energy	-5.0 +/- 1.7
Restraints violation energy	12.6 +/- 11.6
Buried Surface Area	706.8 +/- 40.5
Z-Score	-1.1

Cluster 4

HADDOCK score	-32.7 +/- 5.0
Cluster size	8
RMSD from the overall lowest-energy structure	1.4 +/- 0.2
Van der Waals energy	-15.0 +/- 4.5
Electrostatic energy	-77.2 +/- 6.2
Desolvation energy	-3.3 +/- 0.4
Restraints violation energy	10.8 +/- 12.3
Buried Surface Area	645.7 +/- 13.0
Z-Score	-0.0

MD



Benzamidine

- XP352-BEN

Cluster 7

HADDOCK score	-16.2 +/- 0.8
Cluster size	5
RMSD from the overall lowest-energy structure	0.1 +/- 0.0
Van der Waals energy	-13.4 +/- 0.8
Electrostatic energy	-1.9 +/- 0.6
Desolvation energy	-2.5 +/- 0.1
Restraints violation energy	0.0 +/- 0.0
Buried Surface Area	322.5 +/- 4.0
Z-Score	-2.4

Cluster 9

HADDOCK score	-13.5 +/- 0.2
Cluster size	4
RMSD from the overall lowest-energy structure	0.2 +/- 0.0
Van der Waals energy	-10.6 +/- 0.2
Electrostatic energy	0.3 +/- 1.2
Desolvation energy	-3.0 +/- 0.1
Restraints violation energy	0.8 +/- 0.1
Buried Surface Area	349.9 +/- 7.9
Z-Score	-0.7

Cluster 10

HADDOCK score	-13.2 +/- 0.2
Cluster size	4
RMSD from the overall lowest-energy structure	0.2 +/- 0.0
Van der Waals energy	-10.4 +/- 0.3
Electrostatic energy	-2.6 +/- 1.2
Desolvation energy	-2.6 +/- 0.1
Restraints violation energy	0.5 +/- 0.0
Buried Surface Area	352.1 +/- 2.6
Z-Score	-0.5

- XP273-BEN

Cluster 3

HADDOCK score	-11.8 +/- 0.5
Cluster size	36
RMSD from the overall lowest-energy structure	0.9 +/- 0.0
Van der Waals energy	-10.3 +/- 0.4
Electrostatic energy	2.1 +/- 0.4
Desolvation energy	-1.7 +/- 0.1
Restraints violation energy	0.1 +/- 0.1
Buried Surface Area	266.4 +/- 5.9
Z-Score	-1.1

Cluster 2

HADDOCK score	-11.4 +/- 0.4
Cluster size	40
RMSD from the overall lowest-energy structure	0.9 +/- 0.0
Van der Waals energy	-10.0 +/- 0.3
Electrostatic energy	1.6 +/- 0.4
Desolvation energy	-1.5 +/- 0.1
Restraints violation energy	0.1 +/- 0.1
Buried Surface Area	267.5 +/- 3.2
Z-Score	-0.8

Cluster 1

HADDOCK score	-11.4 +/- 0.2
Cluster size	77
RMSD from the overall lowest-energy structure	0.9 +/- 0.0
Van der Waals energy	-9.9 +/- 0.1
Electrostatic energy	1.9 +/- 0.3
Desolvation energy	-1.7 +/- 0.2
Restraints violation energy	0.0 +/- 0.0
Buried Surface Area	267.9 +/- 4.3
Z-Score	-0.8

- QCL936-BEN

Cluster 4

HADDOCK score	-15.0 +/- 0.0
Cluster size	25
RMSD from the overall lowest-energy structure	0.0 +/- 0.0
Van der Waals energy	-13.4 +/- 0.1
Electrostatic energy	-4.1 +/- 0.2
Desolvation energy	-1.2 +/- 0.0
Restraints violation energy	0.2 +/- 0.1
Buried Surface Area	332.1 +/- 3.1
Z-Score	-1.6

Cluster 3

HADDOCK score	-14.2 +/- 0.2
Cluster size	32
RMSD from the overall lowest-energy structure	0.9 +/- 0.0
Van der Waals energy	-11.8 +/- 0.1
Electrostatic energy	1.0 +/- 0.3
Desolvation energy	-2.4 +/- 0.1
Restraints violation energy	0.0 +/- 0.0
Buried Surface Area	304.6 +/- 1.5
Z-Score	-0.6

Cluster 5

HADDOCK score	-14.1 +/- 0.5
Cluster size	20
RMSD from the overall lowest-energy structure	0.9 +/- 0.0
Van der Waals energy	-11.8 +/- 0.5
Electrostatic energy	0.5 +/- 0.4
Desolvation energy	-2.4 +/- 0.1
Restraints violation energy	0.3 +/- 0.6
Buried Surface Area	307.0 +/- 5.3
Z-Score	-0.5

- ACR157-BEN

Cluster 7

HADDOCK score	-15.1 +/- 0.3
Cluster size	9
RMSD from the overall lowest-energy structure	0.1 +/- 0.0
Van der Waals energy	-14.8 +/- 0.1
Electrostatic energy	-3.0 +/- 0.4
Desolvation energy	-0.0 +/- 0.2
Restraints violation energy	0.0 +/- 0.0
Buried Surface Area	324.1 +/- 0.6
Z-Score	-2.0

Cluster 2

HADDOCK score	-14.7 +/- 0.0
Cluster size	22
RMSD from the overall lowest-energy structure	0.2 +/- 0.0
Van der Waals energy	-14.4 +/- 0.1
Electrostatic energy	-3.2 +/- 0.5
Desolvation energy	0.1 +/- 0.1
Restraints violation energy	0.0 +/- 0.0
Buried Surface Area	327.7 +/- 2.0
Z-Score	-1.4

Cluster 5

HADDOCK score	-13.9 +/- 0.0
Cluster size	16
RMSD from the overall lowest-energy structure	0.5 +/- 0.0
Van der Waals energy	-10.1 +/- 0.2
Electrostatic energy	-1.7 +/- 0.2
Desolvation energy	-3.6 +/- 0.2
Restraints violation energy	0.3 +/- 0.0
Buried Surface Area	271.8 +/- 2.7
Z-Score	-0.3

SKTI

- XP352-SKTI

Cluster 11

HADDOCK score	-77.7 +/- 4.1
Cluster size	5
RMSD from the overall lowest-energy structure	0.9 +/- 0.6
Van der Waals energy	-40.3 +/- 5.9
Electrostatic energy	-216.2 +/- 48.9
Desolvation energy	2.6 +/- 3.1
Restraints violation energy	32.9 +/- 15.2
Buried Surface Area	1248.8 +/- 35.8
Z-Score	-2.3

Cluster 13

HADDOCK score	-61.4 +/- 4.2
Cluster size	4
RMSD from the overall lowest-energy structure	8.1 +/- 0.2
Van der Waals energy	-24.9 +/- 3.0
Electrostatic energy	-221.7 +/- 18.4
Desolvation energy	1.5 +/- 1.4
Restraints violation energy	63.1 +/- 29.3
Buried Surface Area	1237.5 +/- 117.9
Z-Score	-0.6

Cluster 1

HADDOCK score	-60.5 +/- 2.5
Cluster size	72
RMSD from the overall lowest-energy structure	12.4 +/- 0.1
Van der Waals energy	-30.6 +/- 1.6
Electrostatic energy	-169.0 +/- 13.3
Desolvation energy	-0.9 +/- 1.3
Restraints violation energy	47.8 +/- 26.0
Buried Surface Area	1111.2 +/- 42.7
Z-Score	-0.4

- XP273-SKTI

Cluster 2

HADDOCK score	-101.4 +/- 0.6
Cluster size	65
RMSD from the overall lowest-energy structure	1.7 +/- 1.0
Van der Waals energy	-45.5 +/- 0.8
Electrostatic energy	-175.4 +/- 9.4
Desolvation energy	-26.5 +/- 3.5
Restraints violation energy	56.1 +/- 33.7
Buried Surface Area	1540.3 +/- 73.0
Z-Score	-2.0

Cluster 1

HADDOCK score	-86.7 +/- 1.0
Cluster size	68
RMSD from the overall lowest-energy structure	13.2 +/- 0.2
Van der Waals energy	-45.0 +/- 5.0
Electrostatic energy	-93.4 +/- 16.2
Desolvation energy	-25.9 +/- 1.5
Restraints violation energy	29.0 +/- 30.9
Buried Surface Area	1177.3 +/- 34.9
Z-Score	-0.5

Cluster 3

HADDOCK score	-78.3 +/- 0.8
Cluster size	20
RMSD from the overall lowest-energy structure	10.0 +/- 0.5
Van der Waals energy	-39.9 +/- 4.1
Electrostatic energy	-68.6 +/- 3.9
Desolvation energy	-28.7 +/- 1.7
Restraints violation energy	40.9 +/- 22.4
Buried Surface Area	1085.9 +/- 29.9
Z-Score	0.4

- QCL936-SKTI

Cluster 2

HADDOCK score	-113.8 +/- 6.6
Cluster size	37
RMSD from the overall lowest-energy structure	0.9 +/- 0.5
Van der Waals energy	-75.5 +/- 5.7
Electrostatic energy	-131.0 +/- 22.3
Desolvation energy	-13.8 +/- 3.9
Restraints violation energy	16.6 +/- 12.5
Buried Surface Area	1962.7 +/- 85.7
Z-Score	-2.0

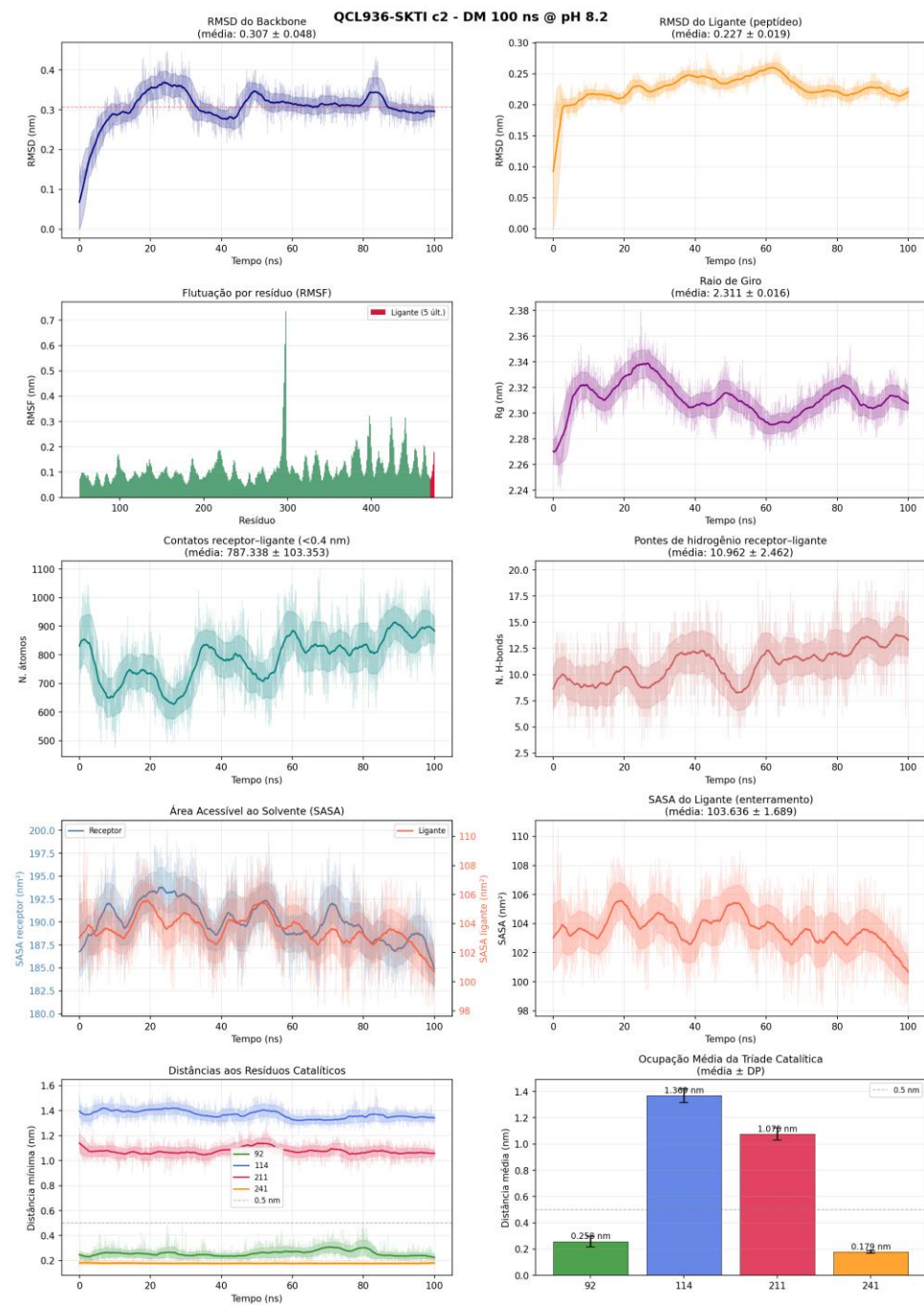
Cluster 1

HADDOCK score	-92.9 +/- 1.4
Cluster size	80
RMSD from the overall lowest-energy structure	12.2 +/- 0.4
Van der Waals energy	-48.8 +/- 5.5
Electrostatic energy	-140.8 +/- 25.4
Desolvation energy	-16.8 +/- 1.2
Restraints violation energy	8.4 +/- 2.3
Buried Surface Area	1669.4 +/- 45.2
Z-Score	-0.8

Cluster 4

HADDOCK score	-79.9 +/- 4.6
Cluster size	14
RMSD from the overall lowest-energy structure	4.1 +/- 0.3
Van der Waals energy	-48.7 +/- 1.0
Electrostatic energy	-159.0 +/- 11.3
Desolvation energy	-3.5 +/- 1.5
Restraints violation energy	41.0 +/- 8.3
Buried Surface Area	1590.0 +/- 77.1
Z-Score	-0.1

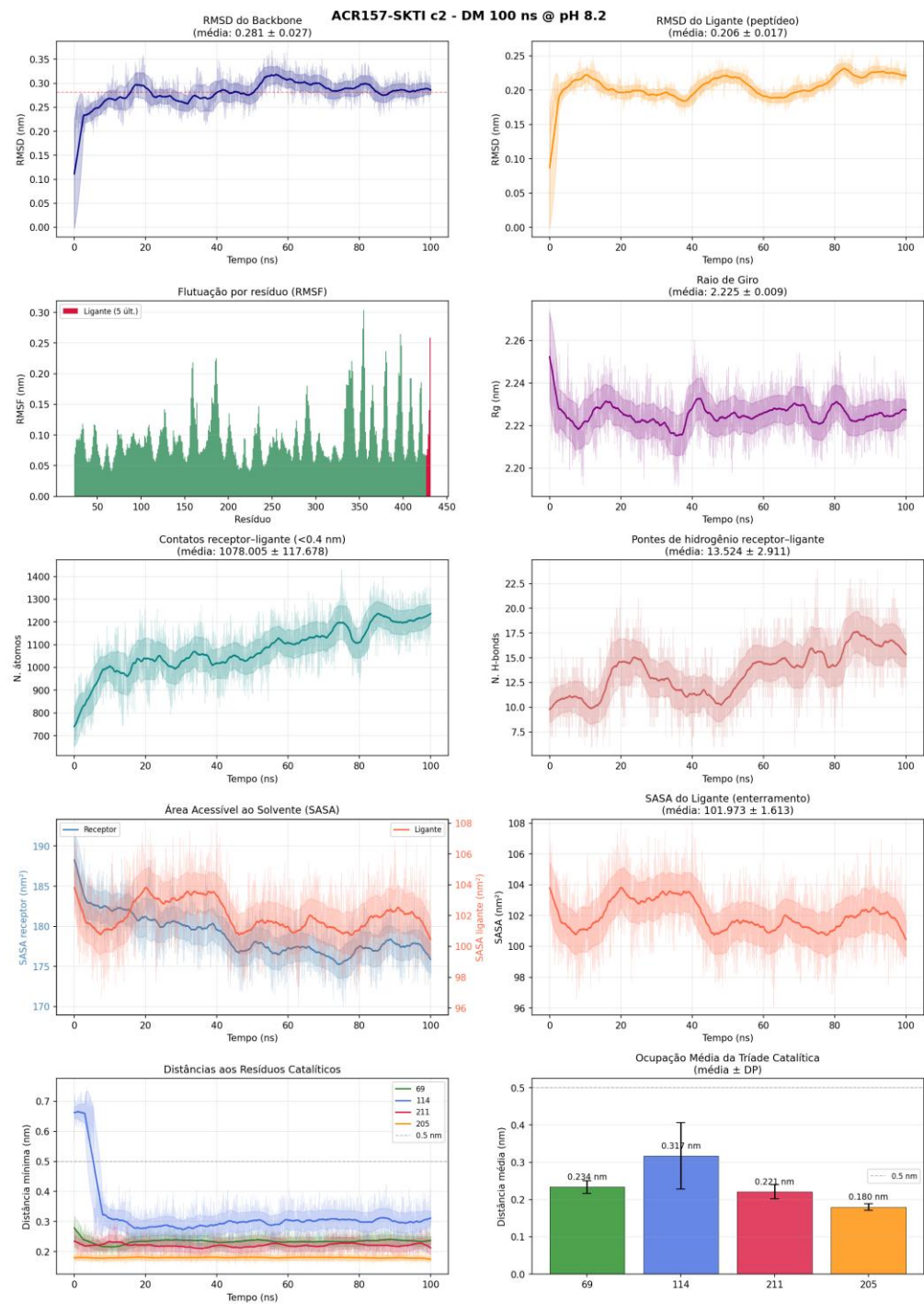
MD



- ACR157-SKTI

Cluster 2		Cluster 4		Cluster 1	
HADDOCK score	-128.3 +/- 3.0	HADDOCK score	-77.2 +/- 5.4	HADDOCK score	-68.2 +/- 1.2
Cluster size	57	Cluster size	7	Cluster size	68
RMSD from the overall lowest-energy structure	1.1 +/- 1.1	RMSD from the overall lowest-energy structure	16.1 +/- 0.1	RMSD from the overall lowest-energy structure	12.7 +/- 0.6
Van der Waals energy	-66.0 +/- 1.9	Van der Waals energy	-38.2 +/- 5.4	Van der Waals energy	-30.4 +/- 4.5
Electrostatic energy	-209.3 +/- 7.4	Electrostatic energy	-113.2 +/- 13.3	Electrostatic energy	-134.8 +/- 21.3
Desolvation energy	-22.2 +/- 2.5	Desolvation energy	-23.9 +/- 1.6	Desolvation energy	-14.6 +/- 1.9
Restraints violation energy	17.3 +/- 12.1	Restraints violation energy	76.1 +/- 22.2	Restraints violation energy	36.9 +/- 22.3
Buried Surface Area	1758.9 +/- 49.5	Buried Surface Area	1424.3 +/- 115.8	Buried Surface Area	1173.9 +/- 25.9
Z-Score	-2.8	Z-Score	-0.4	Z-Score	0.1

MD



Resíduos catalíticos mapeados

Anticarsia

DN258

S70, D119, A219

G213

DN773

H69, D116, S213

G207

DN1238

H73, D117, S211

G205

DN1441

N76, D119, A216

D210

DN1937

H69, D114, S212

D206

DN2954

H79, D126, S222

D216

DN2981

R64, D111, A213

T207

DN3582

H68, D131, S225

G219

DN3963

H194, D243, S335

D229

DN4002

H194, D243, S334

D229

Spodoptera

ACR157

H69, D114, S211

D205

QCL936

H92, D142, S247

D241

XP273

H83, D132, S234

I229

XP352

H112, D166, A268

D262